

Antonios P. Sarikas

AI for Chemistry & Materials

✉ antonios.sarikas@gmail.com

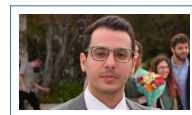
🌐 adosar.github.io/

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🔑 Zv2Uk0AAAAAJ

🆔 0009-0008-8420-927X

3 January 2000



Βελτίων ἢ ἐξ ἐλαττόνων.

💡 Research interests

Topics AI for Materials Discovery • Representation Learning • Foundation Models • Inverse Design

🎓 Education

- 2024–TBA PhD, Chemistry, Department of Chemistry, University of Crete
AI for porous materials
- 2022–2024 MSc, Physical Chemistry, Department of Chemistry, University of Crete
9.86 (Excellent)
- 2018–2022 BSc, Chemistry, Department of Chemistry, University of Crete
8.64 (Excellent)
- First of the class – obtained in 3.5/4 years**

📄 Master thesis

- Title From Potential Energy Surface to Gas Adsorption via Deep Learning
- Description Developed a deep learning framework for predicting gas adsorption in porous materials using 3D energy images.
- Document [Click here to access it](#)

📄 Bachelor thesis

- Title Screening MOFs for H₂ Storage via Machine Learning
- Description Developed a machine learning framework for rapid identification of top-performing MOFs for hydrogen storage.

📢 Scholarships – Awards – Highlights

- 2025 **Excellence Award**, "Professor Zoe Dimitriadis Prize", University of Crete
- 2025 **Best MSc Thesis Award** (3rd place), Federation of European Materials Societies
- 2025 **Postgraduate Fellowship**, Vasileios Apollon Anagnostakis Bequest Fellowship
- 2025 **Best MSc Thesis Award** (1st place), Hellenic Society for the Science and Technology of Condensed Matter
- 2025 **Paper featured in *Nature Portfolio Nobel Prize in Chemistry 2025***. "Gas adsorption meets deep learning: voxelizing the potential energy surface of metal-organic frameworks"
- 2025 **Paper ranked 9th in *Scientific Reports 2024 Materials Science Top 100***. "Gas adsorption meets deep learning: voxelizing the potential energy surface of metal-organic frameworks".

2024 **Postgraduate Fellowship**, Independent Power Transmission Operator S.A. Fellowship
2024 **Postgraduate Fellowship**, Maria Michail Manassaki Bequest Fellowship
2023 **Chatzimarini Award**, Department of Chemistry, University of Crete
2022 **Undergraduate Scholarship**, Stamatiou Foundation
2021 **Undergraduate Scholarship**, Stamatiou Foundation
2020 **Undergraduate Scholarship**, Stamatiou Foundation

Publications

- [1] **Antonios P. Sarikas** and George E. Froudakis. "IntelliPore: A Foundation Model for Gas Adsorption in Porous Materials". In: (2026). doi: 10.26434/chemrxiv.15008693/v1.
- [2] Alexios P. Drosos, **Antonios P. Sarikas**, and George E. Froudakis. "Predicting hydrogen storage in MOFs: Representation matters". In: *International Journal of Hydrogen Energy* 254 (July 2026), p. 156056. doi: 10.1016/j.ijhydene.2026.156056.
- [3] **Antonios P. Sarikas**, Konstantinos Gkagkas, and George E. Froudakis. "RetNeXt: A Pretrained Model for Transfer Learning Across the MOF Adsorption Space". In: *Journal of Chemical Information and Modeling* (Feb. 2026). doi: 10.1021/acs.jcim.5c02698.
- [4] Aydin Ozcan, François-Xavier Coudert, Sven M. J. Rogge, Greta Heydenrych, Dong Fan, **Antonios P. Sarikas**, Seda Keskin, Guillaume Maurin, George E. Froudakis, Stefan Wuttke, and Ilknur Erucar. "Artificial Intelligence Paradigms for Next-Generation Metal–Organic Framework Research". In: *Journal of the American Chemical Society* (June 2025). doi: 10.1021/jacs.5c08214.
- [5] **Antonios P. Sarikas**, Konstantinos Gkagkas, and George E. Froudakis. "Gas adsorption meets geometric deep learning: points, set and match". In: *Scientific Reports* 14.1 (Nov. 2024), p. 27360. doi: 10.1038/s41598-024-76319-8.
- [6] **Antonios P. Sarikas**, Konstantinos Gkagkas, and George E. Froudakis. "Gas adsorption meets deep learning: voxelizing the potential energy surface of metal-organic frameworks". In: *Scientific Reports* 14.1 (Jan. 2024). doi: 10.1038/s41598-023-50309-8.
- [7] **Antonios P. Sarikas**, George S. Fanourgakis, Konstantinos Gkagkas, and George E. Froudakis. "Comparison of machine learning approaches for the identification of top-performing materials for hydrogen storage". In: *Sustainable Chemistry for the Environment* 5 (Mar. 2024), p. 100056. doi: 10.1016/j.scenv.2023.100056.
- [8] **Antonios P. Sarikas**, George S. Fanourgakis, Emmanuel Tylianakis, Konstantinos Gkagkas, and George E. Froudakis. "Comparison of Energy-Based Machine Learning Descriptors for Gas Adsorption". In: *The Journal of Physical Chemistry C* (Oct. 2023). doi: 10.1021/acs.jpcc.3c04223.
- [9] **Antonios P. Sarikas**, George S. Fanourgakis, and George E. Froudakis. "Metal-organic frameworks in the age of machine learning". In: *Reticular Chemistry and Applications*. De Gruyter, Jan. 2023. doi: 10.1515/9781501524721.

Peer-review activities

2026 Reviewer for *npj Artificial Intelligence*
2026 Reviewer for *npj Computational Materials*
2025 Reviewer for *Scientific Reports*
2025 Reviewer for *Communications Chemistry*
2024 Reviewer for *Journal of Cheminformatics*

Software



python™ package for deep learning on porous materials

- Homepage
- Documentation



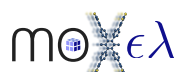
A web app for evaluating pocket-ligand fitness via machine learning

- Click to open the app

Aldorb-online

A web app for predicting properties of porous materials via deep learning

- Click to open the app



python™ package for parallel calculation of energy voxels

- Homepage
- Documentation

RetNeXt

A pretrained model for transfer learning across the MOF adsorption space

- Homepage

RetNet

A 3D ConvNet for Reticular Chemistry

- Homepage

Presentations

- 2025 Memorial seminar in honor of Dr. Chatzimarini, Heraklion
- 2025 Young Scientist Symposium (EuroMOF2025), Heraklion
- 2025 EuroMOF2025, Heraklion
- 2025 FEMS Masters Thesis Award, Granada
- 2025 FEMS EUROMAT 2025, Granada
- 2025 3C Summer School V2, Heraklion
- 2025 HSSTCM Masters Thesis Award, Online
- 2024 COST Action EU4MOFs, Istanbul
- 2023 Memorial seminar in honor of Dr. Chatzimarini, Heraklion
- 2023 1st Mediterranean Conference on Porous Materials, Rethymnon
- 2022 Memorial seminar in honor of Dr. Chatzimarini, Heraklion
- 2022 XXXVI Pan-Hellenic conference on Solid-State Physics and Materials Science, Heraklion

Workshops

- 2024 Computational Materials Science, University of Thessaly, Online
- 2023 Computational Materials Science, National Kapodistrian University of Athens, Online
- 2022 Computational Materials Science, University of Ioannina, Online
- 2021 Computational Materials Science, University of Patras, Online

2021 DCMS Materials 4.0 Summer School, TU Dresden, Online



Computer skills

Operating Systems Fedora Linux, Ubuntu, Windows

Programming Languages Python, Fortran

Markup Languages $\LaTeX$, Markdown

Other Tools Git, Github, Sphinx, Vim, MSOffice



Working experience

2025 Coordinator for 3C Summer School V2, Department of Chemistry, University of Crete

2023, 2025 Teaching assistant on the Laboratory of Physical-Chemistry II, Department of Chemistry, University of Crete

2023, 2024 Teaching assistant on the Laboratory of Physical-Chemistry I, Department of Chemistry, University of Crete



Languages

English Proficiency, University of Michigan

Greek Native language